



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2002

# MATHEMATICS

## EXTENSION 1

2:00am – 4:05 pm  
Friday 30th August 2002

### General Instructions

- Reading time : 5 minutes
- Working time: 2 hours
- Write using blue or black pen
- *Write your name and teacher's name on each answer booklet*
- Board approved calculators may be used
- A table of standard integrals is provided
- Total Marks (84)
- Attempt Questions 1 – 7
- All questions are of equal value

Students are reminded that this is a trial examination only and cannot in any way guarantee the content or the format of the 2002 Mathematics Extension 1 Higher School Certificate examination

### STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:  $\ln x = \log_e x, x > 0$

Total marks (84)

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 marks) Use a SEPARATE writing booklet. Marks

- (a) Factorise  $64 - y^3$ . 1
- (b) Find the coordinates of the point that divides the join of  $A(1, -3)$  and  $B(-5, 7)$  externally in the ratio  $5 : 2$ . 2
- (c) Solve  $\frac{2x}{x-2} > 1$ . 3
- (d) Find the acute angle, to the nearest degree, between the lines  $3x - y + 2 = 0$  and  $x + 2y - 5 = 0$ . 3
- (e) Using the expansion of  $\cos(A + B)$ , find the exact value of  $\cos 75^\circ$  in surd form with a rational denominator. 3

QUESTION 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the general solution of the equation  $\sin^2 x - 2 \cos x + 2 = 0$ , where  $x$  is in radians. 3
- (b) Find  $\int \cos^2 2x \, dx$ . 2
- (c) Use the substitution  $x = u^2 - 1$  for  $u > 0$ , to evaluate  $\int_3^8 \frac{x-1}{\sqrt{x+1}} \, dx$ . 4
- (d) The probability that a student will attend university after doing the HSC is 0.7. In a class of 25 students, what is the probability that at least 23 students will proceed to university after the HSC? 3

QUESTION 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the indefinite integral  $\int \frac{dx}{9+16x^2}$ . 2
- (b) Consider the function  $f(x) = \cos^{-1}(x+2) - \frac{\pi}{2}$ .
- (i) What is the domain of  $y = f(x)$ ? 1
- (ii) Sketch the graph of  $y = f(x)$ . 2
- (c) Consider the function  $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ .
- (i) Prove the derivative of  $f(x)$  is zero. 2
- (ii) What is the domain of  $y = f(x)$ ? 1
- (iii) Find the value of  $f(x)$  over the domain of  $y = f(x)$ . 2
- (d) The velocity  $v$  of a particle moving in a straight line at position  $x$  is given by: 2

$$v = 1 + 2x.$$

Find the acceleration of the particle at position  $x$ .

QUESTION 4 (12 marks) Use a SEPARATE writing booklet.

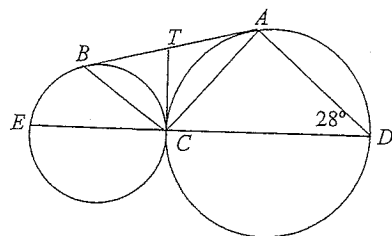
Marks

- (a) Consider the letters of the word MILLER.
- (i) How many arrangements of these letters are possible if the letters are arranged in a straight line? 1
- (ii) What is the probability that the L's will be separated when the letters are arranged in a straight line? 2
- (iii) If the letters are arranged in a circle, how many arrangements are possible? 1
- (iv) If the letters are arranged in a circle, what is the probability the L's will be opposite each other? 2
- (b) Find the term independent of  $x$  in the expansion of  $\left(x - \frac{2}{x^2}\right)^9$ , expressing your answer as an integer. 3
- (c) Prove that  $\binom{n+1}{2} + \binom{n+2}{2}$  is the square of an integer. 3

QUESTION 6 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)

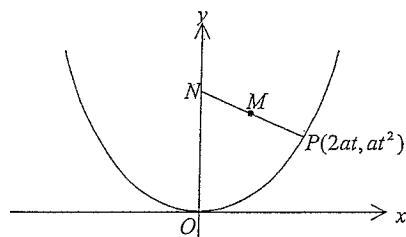


4

$AB$  is a common tangent to the circles, which touch at  $C$ .  $CT$  is the tangent at  $C$ . A line through  $C$  meets the circles at  $D$  and  $E$ .  $\angle ADC = 28^\circ$ .

Find the size of  $\angle ACB$ , giving reasons.

(b)



A normal is drawn to the parabola  $x^2 = 4ay$  at a variable point  $P(2at, at^2)$  on the parabola. The normal meets the  $y$  axis at  $N$ .  $M$  is the midpoint of  $PN$ . The coordinates of  $M$  are  $(at, at^2 + a)$  and  $O$  is the origin.

(i) Show that the equation of the locus of  $M$  is the parabola  $ay = x^2 + a^2$ . 1

(ii) What is the vertex of the parabola  $ay = x^2 + a^2$ ? 1

(iii) What is the domain of this locus? Explain. 1

(c) The rate of change of the amount of pollutant ( $x$  units) in a lake after  $t$  days of rain is given by:  $\frac{dx}{dt} = \frac{1}{4} - \frac{x}{16}$ .

(i) Show that  $x = 4 + Ae^{-\frac{t}{16}}$  satisfies this equation. 2

(ii) If the initial amount of pollutant is 20 units, find the value of  $A$ . 1

(iii) In how many days will the amount of the pollutant drop to 8 units? 2

QUESTION 5 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Two of the roots of the equation  $x^3 + 6x^2 - x + k = 0$  are 2, -3, where  $k$  is a constant.

(i) By considering the sum of the roots of this equation, find the third root. 2

(ii) Hence or otherwise find the value of the constant  $k$ . 2

(b) Use one step of Newton's method to find an approximation to the root of the equation  $x^4 - 10x + 7 = 0$  near  $x = 2$ . 3

(c) A particle moves in Simple Harmonic Motion about an origin  $O$ . The period of motion is 4 seconds and the amplitude is 6 cm.

(i) Write down an expression for  $x$  in terms of  $t$ , when the particle is  $x$  cm from  $O$ ,  $t$  seconds after passing through  $O$ . 2

(ii) Find its speed as it passes through  $O$ . 2

(iii) What is the maximum acceleration of the particle? 1



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**MATHEMATICS**

**EXTENSION 1**

Suggested Solutions

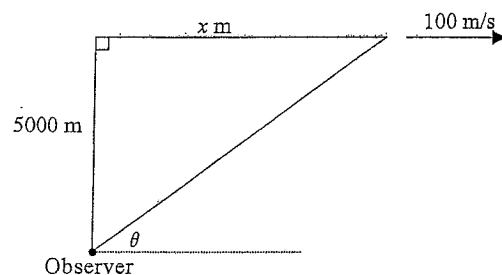
QUESTION 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove by mathematical induction, where  $n$  is a positive integer and  $n \geq 2$ , 5

that  $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$ .

- (b)



At a certain instant, a plane flies overhead at a constant altitude of 5000 metres, at a constant speed of 100 metres per second. When the plane has travelled  $x$  metres from the overhead position, its angle of elevation from the observer is  $\theta$  radians.

- (i) Show that  $\frac{dx}{d\theta} = -\frac{5000}{\sin^2 \theta}$ . 2
- (ii) Hence show that  $\frac{d\theta}{dt} = -\frac{1}{50} \sin^2 \theta$ . 2
- (iii) Find the rate at which the angle of elevation is changing:
- ( $\alpha$ ) when the plane is overhead. 1
- ( $\beta$ ) 50 seconds after the plane is overhead. 2

End of Examination

| Mathematics Extension 1: Question 1                                                                                                                                                                                                                                                                                                                                       |               |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                       | Marks Awarded | Marker's Comments |
| (a) $64 - y^3 = (4 - y)(16 + 4y + y^2)$ (1)                                                                                                                                                                                                                                                                                                                               |               |                   |
| (b) $A(1, -3) \quad B(-5, 7) \quad S: -2$<br>$\left( \frac{5 \times (-5) + (-2) \times 1}{5 + (-2)}, \frac{5 \times 7 + (-2) \times (-3)}{5 + (-2)} \right)$<br>$(-9, 13\frac{2}{3})$ (2)                                                                                                                                                                                 |               |                   |
| (c) $\frac{2x}{x-2} > 1$<br>$2x(x-2) > (x-2)^2$<br>$2x^2 - 4x > x^2 - 4x + 4$<br>$x^2 > 4$<br>$x < -2, x > 2$ (3)                                                                                                                                                                                                                                                         |               |                   |
| (d) $3x - y + 2 = 0: m = 3$<br>$x + 2y - 5 = 0: m = -\frac{1}{2}$<br>$\tan \theta = \left  \frac{3 - (-\frac{1}{2})}{1 + 3 \times (-\frac{1}{2})} \right $<br>$= 7$<br>$\theta = 82^\circ$ (nearest degree) (3)                                                                                                                                                           |               |                   |
| (e) $\cos(A+B) = \cos A \cos B - \sin A \sin B$<br>$\cos 75^\circ = \cos(45^\circ + 30^\circ)$<br>$= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$<br>$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$<br>$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$<br>$= \frac{\sqrt{6}-\sqrt{2}}{4}$ (1.5) |               |                   |

| Mathematics Extension 1: Question 2                                                                                                                                                                                                                                                                                               |               |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| Suggested Solutions                                                                                                                                                                                                                                                                                                               | Marks Awarded | Marker's Comments |
| (a) $\sin^2 x - 2 \cos x + 2 = 0$<br>$1 - \cos^2 x - 2 \cos x + 2 = 0$<br>$\cos^2 x + 2 \cos x - 3 = 0$<br>$(\cos x + 3)(\cos x - 1) = 0$<br>$\cos x = -3 : \text{no solution}$<br>$\cos x = 1 : x = 2n\pi$<br>where $n$ is an integer. (3)                                                                                       |               |                   |
| (b) $\int \cos^2 2x \, dx = \int \frac{1}{2} (1 + \cos 4x) \, dx$<br>$= \frac{1}{2} (x + \frac{1}{4} \sin 4x) + C$ (2)                                                                                                                                                                                                            |               |                   |
| (c) $\int_3^8 \frac{x-1}{\sqrt{x+1}} \, dx$<br>$= \int_2^3 \frac{u^2-2}{u} \times 2u \, du$<br>$= \int_2^3 (2u^2-4) \, du$<br>$= \left[ \frac{2}{3} u^3 - 4u \right]_2^3$<br>$= (18-12) - (5\frac{1}{3}-8)$<br>$= 8\frac{2}{3}$<br>$x = u^2 - 1$<br>$dx = 2u \, du$<br>$x = 3 \Rightarrow u = 2$<br>$x = 8 \Rightarrow u = 3$ (4) |               |                   |
| (d) $P(\text{at least 23 students})$<br>$= P(23) + P(24) + P(25)$<br>$= \binom{25}{23} (0.7)^{23} (0.3)^2 + \binom{25}{24} (0.7)^{24} (0.3) + (0.7)^{25}$<br>$= 0.00896$ (3)                                                                                                                                                      |               |                   |

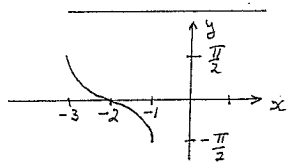
# Mathematics Extension 1: Question 3

Suggested Solutions

$$\begin{aligned}(a) \int \frac{dx}{9+16x^2} &= \int \frac{dx}{16\left(\frac{9}{16}+x^2\right)} \\ &= \frac{1}{16} \times \frac{1}{\frac{3}{4}} \tan^{-1} \frac{2x}{\frac{3}{4}} + C \\ &= \frac{1}{12} \tan^{-1} \frac{4x}{3} + C\end{aligned}$$

OR Use substitution  $u = 4x$ .

$$\begin{aligned}(b) \quad f(x) &= \cos^{-1}(x+2) - 1 \\ (i) \text{ Domain: } &-1 \leq x+2 \leq 1 \\ \text{i.e. } &-3 \leq x \leq -1\end{aligned}$$



$$\begin{aligned}(c) \quad f(x) &= \tan^{-1}x + \tan^{-1}\frac{1}{x} \\ (i) \quad f'(x) &= \frac{1}{1+x^2} + \frac{1}{1+(\frac{1}{x})^2} \times \left(-\frac{1}{x^2}\right) \\ &= \frac{1}{1+x^2} - \frac{1}{x^2+1} \\ &= 0\end{aligned}$$

(ii) Domain: all  $x$ , except  $x=0$

$$\begin{aligned}(iii) \text{ For } x > 0: f(1) &= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \\ \text{For } x < 0: f(-1) &= -\frac{\pi}{4} + \left(-\frac{\pi}{4}\right) = -\frac{\pi}{2} \\ \therefore f(x) &= \frac{\pi}{2} \text{ for } x > 0; f(x) = -\frac{\pi}{2} \text{ for } x < 0.\end{aligned}$$

$$\begin{aligned}(d) \quad v &= 1+2x \\ a &= \frac{dv}{dx} \left(\frac{1}{2}v^2\right) \\ &= \frac{dv}{dx} \cdot \frac{1}{2} (1+2x)^2 \\ &= \frac{1}{2} \times 2(1+2x) \times 2 \\ &= 2(1+2x).\end{aligned}$$

# Mathematics Extension 1: Question 4

Suggested Solutions

Marks  
Awarded

Marker's Comments

(a) MILLER.

$$(i) \text{ No. of arrangements} = \frac{6!}{2!} = 360 \quad (1)$$

$$\begin{aligned}(ii) \text{ Prob (L's separated)} \\ &= 1 - P(\text{L's together}) \\ &= 1 - \frac{5!}{360} \\ &= \frac{2}{3} \quad (2)\end{aligned}$$

$$(iii) \text{ No. of arrangements} = \frac{5!}{2!} = 60 \quad (1)$$

$$\begin{aligned}(iv) P(\text{L's opposite}) &= \frac{1}{5} \\ \text{or } \frac{4!}{5!} &= \frac{1}{5} \quad (2)\end{aligned}$$

$$(b) \left(x - \frac{2}{x^2}\right)^9.$$

$$\begin{aligned}\text{Term indep. of } x &= \binom{9}{3} x^6 \left(-\frac{2}{x^2}\right)^3 \\ (\text{by inspection}) &= 504 \times (-8) \times x^0 \\ &= -672\end{aligned}$$

$$\begin{aligned}\text{OR } T_{r+1} &= \binom{9}{r} x^{9-r} \left(-\frac{2}{x^2}\right)^r \\ &= \binom{9}{r} (-2)^r x^{9-3r}\end{aligned} \quad (3)$$

$$\begin{aligned}(c) \binom{n+1}{2} + \binom{n+2}{2} &= \frac{(n+1)n}{2} + \frac{(n+2)(n+1)}{2} \\ &= \frac{n+1}{2} [n+n+2] \\ &= \frac{n+1}{2} \times 2(n+1) \\ &= (n+1)^2\end{aligned}$$

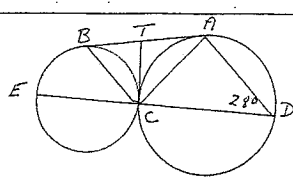
which is the square of an integer.

(3)

# Mathematics Extension 1: Question 5

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks Awarded | Marker's Comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| <p>(a) <math>x^3 + 6x^2 - x + k = 0</math></p> <p>(i) Let the third root be <math>\alpha</math>.<br/> <math>2 + (-3) + \alpha = -6</math><br/> <math>\alpha = -5</math> (2)</p> <p>(ii) By product of roots:<br/> <math>2 \times (-3) \times (-5) = -k</math><br/> <math>k = -30</math> (2)</p>                                                                                                                                                                                                                                                                                                            |               |                   |
| <p>(b) <math>f(x) = x^4 - 10x + 7</math><br/> <math>f'(x) = 4x^3 - 10</math><br/> <math>x_1 = 2 - \frac{f(2)}{f'(2)}</math><br/> <math>= 2 - \frac{3}{22}</math><br/> <math>= 1.86</math> (2 d.p.) (3)</p>                                                                                                                                                                                                                                                                                                                                                                                                 |               |                   |
| <p>(c) Period = 4 sec, Amplitude = 6 cm.</p> <p>(i) <math>T = \frac{2\pi}{n} = 4 \therefore n = \frac{\pi}{2}</math>.<br/> <math>x = 6 \sin \frac{\pi}{2} t</math>. (2)</p> <p>(ii) <math>v = 6 \times \frac{\pi}{2} \cos \frac{\pi}{2} t</math>.<br/>         When <math>t=0</math>, <math>v = 3\pi</math>.<br/>         Speed is <math>3\pi</math> cm/s at 0. (2)</p> <p>(iii) <math>a = -3\pi \times \frac{\pi}{2} \sin \frac{\pi}{2} t</math><br/> <math>= -\frac{3\pi^2}{2} \sin \frac{\pi}{2} t</math><br/>         Maximum acceleration is <math>\frac{3\pi^2}{2}</math> cm/sec<sup>2</sup> (1)</p> |               |                   |

# Mathematics Extension 1: Question 6

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks Awarded | Marker's Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| <p>(a) </p> <p><math>\angle TAC = 28^\circ</math> (alternate seg. theorem)<br/> <math>\angle TCA = 28^\circ</math> (TA = TC: tangents from an external point are equal)<br/> <math>\angle BTC = 56^\circ</math> (exterior angle of <math>\triangle TAC</math>).<br/>         But <math>TB = TC</math> (tangents from ext. point)<br/> <math>\therefore \angle TCB = \frac{1}{2}(180^\circ - 56^\circ)</math> (angle sum of isos. <math>\triangle TBC</math>)<br/> <math>= 62^\circ</math><br/> <math>\therefore \angle ACB = 28^\circ + 62^\circ = 90^\circ</math>. (4)</p> |               |                   |
| <p>(b)(i) Coord of M. <math>x = at</math>, <math>y = at^2 + a</math>.<br/> <math>t = \frac{x}{a} \therefore y = a \times \frac{x^2}{a^2} + a</math><br/> <math>ay = x^2 + a^2</math> (1)</p> <p>(ii) <math>x^2 = a(y - a)</math> Vertex is <math>(0, a)</math>. (1)</p> <p>(iii) Domain - all <math>x</math> except <math>x=0</math>, because normal at <math>(0,0)</math> does not cut the <math>y</math>-axis at one point - it is the <math>y</math>-axis (1)</p>                                                                                                                                                                                           |               |                   |
| <p>(c) <math>\frac{dx}{dt} = \frac{1}{4} - \frac{x}{16}</math>.</p> <p>(i) <math>x = 4 + Ae^{-\frac{t}{16}}</math><br/> <math>\frac{dx}{dt} = -\frac{1}{16} Ae^{-\frac{t}{16}} = -\frac{1}{16}(x - 4) = \frac{1}{4} - \frac{x}{16}</math>. (2)</p> <p>(ii) <math>t=0, x=20</math>: <math>20 = 4 + Ae^0 \therefore A = 16</math> (1)</p> <p>(iii) <math>x = 4 + 16e^{-\frac{t}{16}}</math><br/>         When <math>x=8</math>, <math>8 = 4 + 16e^{-\frac{t}{16}}</math><br/> <math>e^{-\frac{t}{16}} = \frac{1}{4}</math>, <math>e^{\frac{t}{16}} = 4</math><br/> <math>t = 16 \ln 4 = 22.18</math><br/>         Approx 22 days. (2)</p>                        |               |                   |



Mathematics Extension 1: Question 7(a)

Suggested Solutions

Marks  
Awarded

Marker's Comments

(a)

Prove  $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$  for  $n \geq 2$ .

When  $n=2$ , LHS =  $1 - \frac{1}{2^2} = \frac{3}{4}$ .

RHS =  $\frac{2+1}{2 \times 2} = \frac{3}{4}$

$\therefore$  it is true for  $n=2$ .

Assume it is true for  $n=k$ :

i.e. assume  $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}$ .

When  $n=k+1$ ,

LHS =  $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{k^2}\right)\left(1 - \frac{1}{(k+1)^2}\right)$

=  $\frac{k+1}{2k} \left(1 - \frac{1}{(k+1)^2}\right)$  by assumption

=  $\frac{k+1}{2k} \times \frac{(k+1)^2 - 1}{(k+1)^2}$

=  $\frac{k^2 + 2k + 1 - 1}{2k(k+1)}$

=  $\frac{k^2 + 2k}{2k(k+1)}$

=  $\frac{k(k+2)}{2k(k+1)}$

=  $\frac{k+2}{2(k+1)}$

=  $\frac{n+1}{2n}$  when  $n=k+1$ .

$\therefore$  if it is true for  $n=k$ ,  
then it is true for  $n=k+1$ .

Since it is true for  $n=2$ ,  
it is true for  $n=3, n=4, \dots$

i.e. it is true for all positive  
integers  $n \geq 2$

(5)

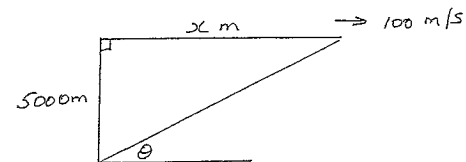
Mathematics Extension 1: Question 7(b)

Suggested Solutions

Marks  
Awarded

Marker's Comments

(b)



$$(i) \quad \frac{x}{5000} = \tan(90^\circ - \theta)$$

$$x = 5000 \cot \theta$$

$$\frac{dx}{d\theta} = -5000 \operatorname{cosec}^2 \theta$$

$$= -\frac{5000}{\sin^2 \theta}$$

(2)

$$(ii) \quad \frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$100 = -\frac{5000}{\sin^2 \theta} \times \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = -\frac{100 \sin^2 \theta}{5000}$$

$$= -\frac{1}{50} \sin^2 \theta$$

(2)

(iii) (α) When overhead,  $\theta = 90^\circ$ ,

$$\frac{d\theta}{dt} = -\frac{1}{50} \sin^2 90^\circ$$

$$= -\frac{1}{50}$$

Angle is changing at  $\frac{1}{50}$  rad/sec. (1)

(β) When  $t=50$ ,  $x = 100 \times 50$   
 $= 5000$

$$\theta = \frac{\pi}{4}$$

$$\frac{d\theta}{dt} = -\frac{1}{50} \sin^2\left(\frac{\pi}{4}\right)$$

$$= -\frac{1}{100}$$

Angle is changing at  $\frac{1}{100}$  rad/sec. (2)